

# Solutions to Problem Set 1

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## Problem 1

1. Constraint  $B_{T+1} \geq 0$  is not binding in this model because the agent's utility converges to  $-\infty$  when  $B_{T+1} \leq 0$ . Therefore, she will always choose a positive bequest.
2. The FOCs with respect to  $C_t$ ,  $B_{t+1}$  and  $B_{T+1}$  are respectively:

$$\begin{aligned}\beta^t \frac{1}{C_t} &= \lambda_t \\ \lambda_t &= R\lambda_{t+1} \\ \beta^T \eta \frac{1}{B_{T+1}} &= \lambda_T\end{aligned}$$

Substitute out the Lagrange multipliers to obtain the Euler equations for  $t < T$  and the transversality condition for  $t = T$ :

$$C_{t+1} = \beta RC_t, \quad B_{T+1} = \eta C_T.$$

The latter condition says that in the last period, the agent equalizes the marginal utility from spending money on consumption and from leaving it as a bequest.

3. Following the lecture notes, iterate the budget constraint to get

$$\sum_{t=0}^T R^{-t} (C_t - Y) + R^{-T} B_{T+1} = 0.$$

Substitute consumption  $C_t = (\beta R)^t C_0$  and bequest  $B_{T+1} = \eta (\beta R)^T C_0$  from the opti-

mality conditions and solve for  $C_0$ :

$$C_0 = \frac{1 - \beta}{1 + \eta\beta^T - (1 + \eta)\beta^{T+1}} \frac{R - R^{-T}}{R - 1} Y.$$

4. In the limit  $T \rightarrow \infty$ , the optimal bequest  $B_{T+1} \rightarrow \eta Y$ . Intuitively, utility from bequest is isomorphic to utility from consumption in a period far in the future. When  $\beta R = 1$ , the agent perfectly smooths consumption in time and also leaves a positive bequest.

## Problem 2

1. By the L'Hopital's rule,

$$\lim_{\sigma \rightarrow 1} \frac{C_t^{1-\sigma} - 1}{1 - \sigma} = \lim_{\sigma \rightarrow 1} \frac{e^{(1-\sigma) \log C_t} - 1}{1 - \sigma} = \lim_{\sigma \rightarrow 1} \frac{-e^{(1-\sigma) \log C_t} \log C_t}{-1} = \log C_t.$$

2. The Hamiltonian

$$\mathcal{H}_t = e^{-\rho t} \log C_t + \lambda_t (rB_t - C_t)$$

and the principle of maximum

$$\begin{aligned} 0 &= \frac{\partial \mathcal{H}_t}{\partial C_t} = e^{-\rho t} C_t^{-1} - \lambda_t \\ -\dot{\lambda}_t &= \frac{\partial \mathcal{H}_t}{\partial B_t} = r\lambda_t \end{aligned}$$

allow to obtain the Euler equation

$$C_t = C_0 e^{(r-\rho)t} = C_0$$

and the transversality condition  $B_T = 0$ .

3. Following the lecture notes, obtain

$$e^{-rT} B_T = B_0 - \int_0^T e^{-rs} C_s ds.$$

According to the transversality condition, the left-hand side of the equation is zero.

Substitute in the Euler equation and integrate

$$C_t = \frac{\rho B_0}{1 - e^{-\rho T}}.$$

4. Substitute the optimal consumption path into the objective function and integrate:

$$\int_0^T e^{-\rho t} \log \frac{\rho B_0}{1 - e^{-\rho T}} dt = \frac{1}{\rho} (1 - e^{-\rho T}) \left( \log \rho B_0 - \log(1 - e^{-\rho T}) \right).$$

Take the first-order condition wrt  $T$ :

$$\rho e^{-\rho T} \log \rho B_0 - \rho e^{-\rho T} \log(1 - e^{-\rho T}) - \rho e^{-\rho T} = 0.$$

It follows that the optimal  $T$  is given by

$$T = \begin{cases} -\frac{1}{\rho} \log(1 - \rho e^{-1} B_0), & \text{if } \rho e^{-1} B_0 < 1 \\ +\infty, & \text{otherwise} \end{cases}.$$

Notice that the period of life is longer when initial wealth  $B_0$  is high and the discount factor  $\rho$  is smaller.

5. Start with the optimal path of consumption and the optimal  $T$ . Suppose that the agent increases  $T$  by  $\Delta$ . Given a constant consumption in time, the agent spends  $C_T \Delta$  over this period, which requires a proportionate increase in  $B_T$ . According to the budget constraint  $e^{-rT} B_T = B_0 - \frac{1}{r} (1 - e^{-rT}) C_T$ , this change translates in cutting consumption in all periods by  $dC_t = -\frac{r C_T \Delta}{e^{rT} - 1}$ . Given  $r = \rho$ , the net change in the utility is equal

$$e^{-\rho T} u(C_T) \Delta + \int_0^T e^{-\rho t} dt u'(C_T) dC_T = e^{-\rho T} \left( \log C_T - 1 \right) \Delta$$

and under the interior solution, has to be equal to zero to insure the optimality of  $T$ . It follows that  $C_T = \frac{\rho B_0}{1 - e^{-\rho T}} = e$ , which can be solved for  $T$ .

## Problem 3

1. Write the Lagrangian

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left[ A_t K_t^\alpha - I_t - \frac{\phi}{2} \frac{I_t^2}{K_t} \right] + \sum_{t=0}^{\infty} \lambda_t \left( (1 - \delta) K_t + I_t - K_{t+1} \right)$$

and take the first-order conditions

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial I_t} &= -\beta^t \left( 1 + \phi \frac{I_t}{K_t} \right) + \lambda_t = 0, \\ \frac{\partial \mathcal{L}}{\partial K_{t+1}} &= \beta^{t+1} \left( \alpha A_{t+1} K_{t+1}^{\alpha-1} + \frac{\phi}{2} \frac{I_{t+1}^2}{K_{t+1}^2} \right) - \lambda_t + (1 - \delta) \lambda_{t+1} = 0. \end{aligned}$$

2. In a finite-period version of the model, we need to add that the firm cannot leave a negative stock of capital, i.e.  $K_{T+1} \geq 0$ . Denote the Lagrange multiplier on this constraint with  $\mu$  and take the optimality condition wrt  $K_{T+1}$ :

$$\frac{\partial \mathcal{L}}{\partial K_{T+1}} = -\lambda_T + \mu = 0.$$

It follows that the complementary slackness condition can be written as

$$\lambda_T K_{T+1} = 0.$$

By analogy, the transversality condition in an infinite-horizon model is

$$\lim_{t \rightarrow \infty} \lambda_t K_{t+1} = 0.$$

3. Combine the first-order conditions into

$$1 + \phi \frac{I_t}{K_t} = \beta \left( \alpha A_{t+1} K_{t+1}^{\alpha-1} + \frac{\phi}{2} \frac{I_{t+1}^2}{K_{t+1}^2} \right) + \beta(1 - \delta) \left( 1 + \phi \frac{I_{t+1}}{K_{t+1}} \right).$$

Suppose that the firm increases investment in period  $t$  by  $\Delta$ , which requires spending  $\left( 1 + \phi \frac{I_t}{K_t} \right) \Delta$ . This increases capital next period by  $\Delta$  and brings additional profits of  $\alpha A_{t+1} K_{t+1}^{\alpha-1} \Delta$ . To return back to the original trajectory and keep  $K_{t+2}$  unchanged, the firm has to lower investment at  $t + 1$  by  $(1 - \delta) \Delta$ . Given changes in  $I_{t+1}$  and  $K_{t+1}$ , the spendings at  $t + 1$  change by  $\frac{\phi}{2} \frac{I_{t+1}^2}{K_{t+1}^2} \Delta + \left( 1 + \phi \frac{I_{t+1}}{K_{t+1}} \right) (1 - \delta) \Delta$ . Combining all these

benefits and costs, we get the optimality condition from above.

4. Define  $\beta \equiv e^{-r}$  and note that capital is a stock variable, while investment, depreciation, and profits are flows. It follows that

$$K_{t+\Delta} = (1 - \delta\Delta)K_t + I_t\Delta.$$

Rearranging terms, we get

$$K_{t+\Delta} - K_t = (I_t - \delta K_t)\Delta$$

and taking the limit  $\Delta \rightarrow 0$ ,

$$\bar{K}_t = I_t - \delta K_t, \quad \text{where } \bar{K}_t \equiv \frac{K_{t+\Delta} - K_t}{\Delta}.$$

In turn, the objective is

$$\int_0^\infty e^{-rt} \left( A_t K_t^\alpha - I_t - \frac{\phi}{2} \frac{I_t^2}{K_t} \right) dt.$$

5. Let  $\lambda_t$  denote the co-state. Then the Hamiltonian is given by

$$\mathcal{H}_t = e^{-rt} \left( A_t K_t^\alpha - I_t - \frac{\phi}{2} \frac{I_t^2}{K_t} \right) + \lambda_t (I_t - \delta K_t).$$

Since  $I_t$  is a control variable and  $K_t$  is a state variable, the optimality conditions are

$$\begin{aligned} 0 &= \frac{\partial \mathcal{H}_t}{\partial I_t} = -e^{-rt} \left( 1 + \phi \frac{I_t}{K_t} \right) + \lambda_t, \\ -\dot{\lambda}_t &= \frac{\partial \mathcal{H}_t}{\partial K_t} = e^{-rt} \left( \alpha A_t K_t^{\alpha-1} + \frac{\phi}{2} \frac{I_t^2}{K_t^2} \right) - \delta \lambda_t. \end{aligned}$$

6. Indeed, the optimality conditions in discrete time can be written for an arbitrary period length as

$$\begin{aligned} -\beta^t \left( 1 + \phi \frac{I_t}{K_t} \right) \Delta + \lambda_t \Delta &= 0, \\ \beta^{t+\Delta} \left( \alpha A_{t+\Delta} K_{t+\Delta}^{\alpha-1} + \frac{\phi}{2} \frac{I_{t+\Delta}^2}{K_{t+\Delta}^2} \right) \Delta - \delta \lambda_{t+\Delta} \Delta &= -(\lambda_{t+\Delta} - \lambda_t) \end{aligned}$$

Use  $\beta \equiv e^{-r}$  and take the limit  $\Delta \rightarrow 0$  to get the continuous-time optimality conditions.